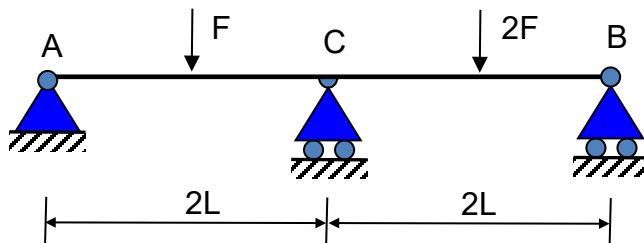


Tutorial 2 – Internal actions

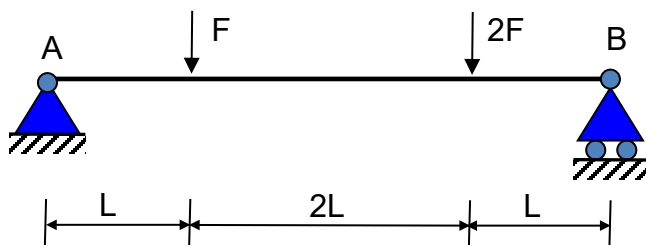
Calculation of internal actions of planar structures and related diagrams.

- Students are expected to actively participate to tutorials, whose aim is to help students in understanding theory and preparing them for the written exam.
- Examples will be partly solved by the tutor, partly by the students themselves who can always ask for a little help when difficulties or doubts arise.
- Examples must be solved in an structured manner, drawing the graphs in scale and writing the basic step for of the solution and the final results.
- It is advisable that students work in small groups (2 or 3 persons).

Example 1:

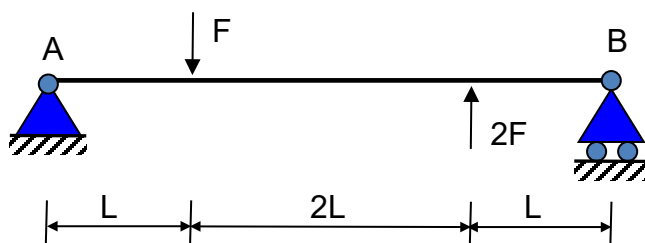


Example 2:



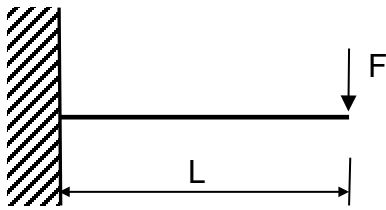
$$\begin{aligned} H_A &= 0 \\ V_A &= \frac{5}{4}F \\ V_B &= \frac{7}{4}F \end{aligned}$$

Example 3:



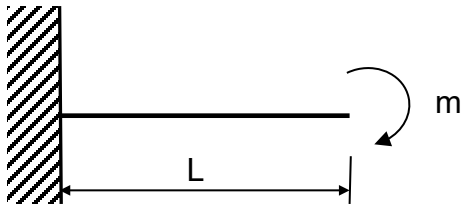
$$\begin{aligned} H_A &= 0 \\ V_A &= \frac{F}{4} \\ V_B &= -\frac{5}{4}F \end{aligned}$$

Example 4:



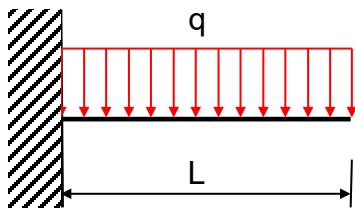
$$\begin{aligned} H_A &= 0 \\ V_A &= F \\ C_A &= -FL \end{aligned}$$

Example 5:



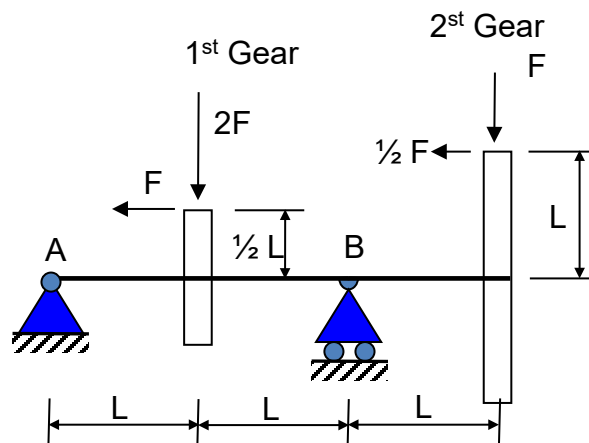
$$\begin{aligned} H_A &= 0 \\ V_A &= 0 \\ C_A &= -m \end{aligned}$$

Example 6:



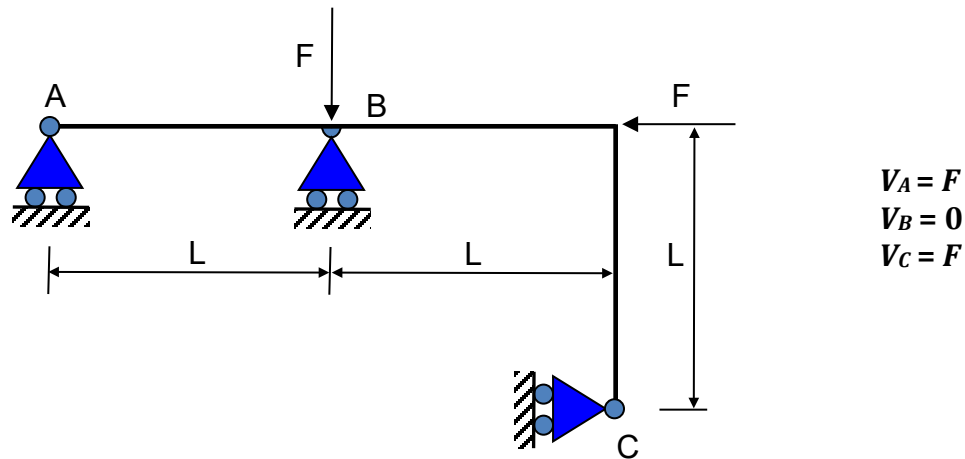
$$\begin{aligned} H_A &= 0 \\ V_A &= qL \\ C_A &= -q \frac{L^2}{2} \end{aligned}$$

Example 7:

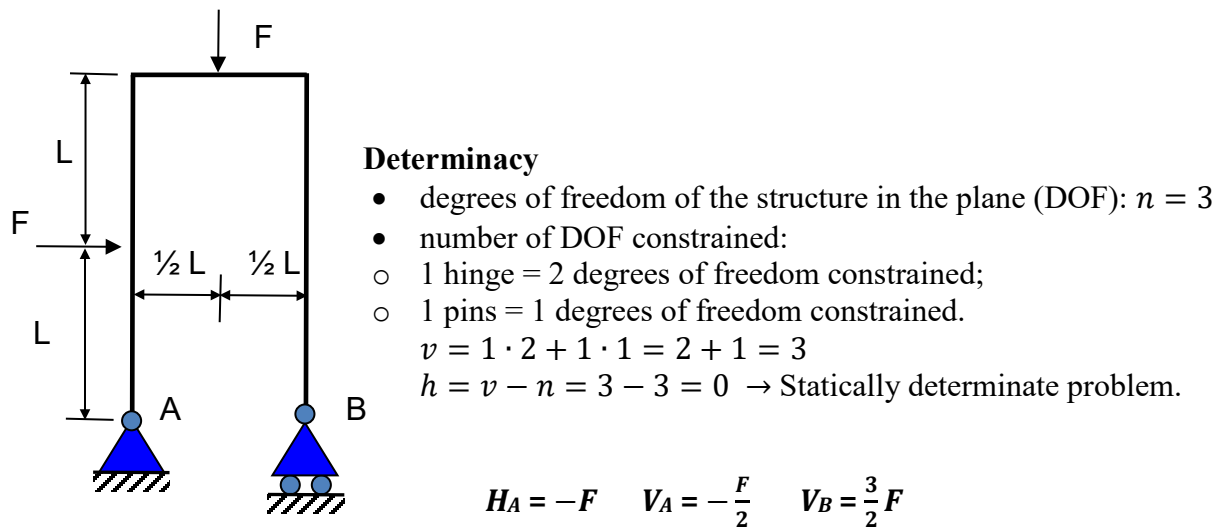


$$\begin{aligned} H_A &= \frac{3}{2}F \\ V_A &= F \\ V_B &= 2F \end{aligned}$$

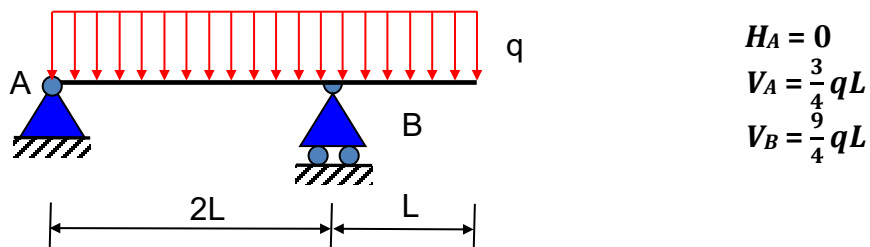
Example 8:



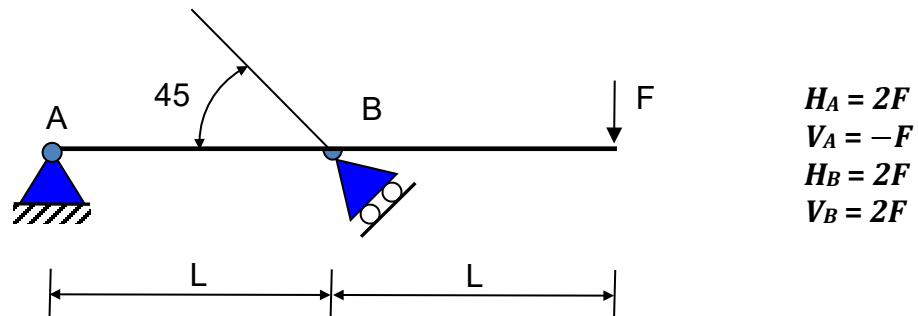
Example 9:



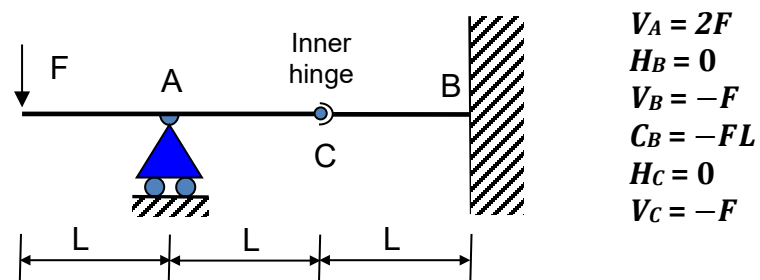
Example 10:



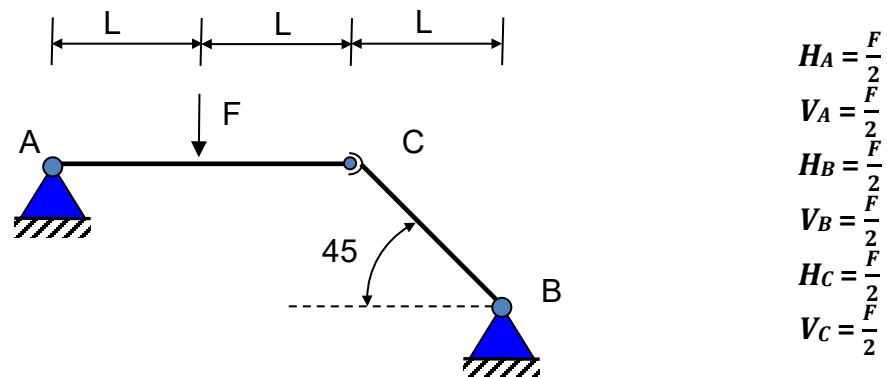
Example 11:



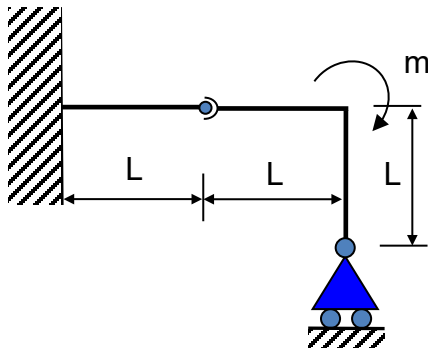
Example 12:



Example 13:



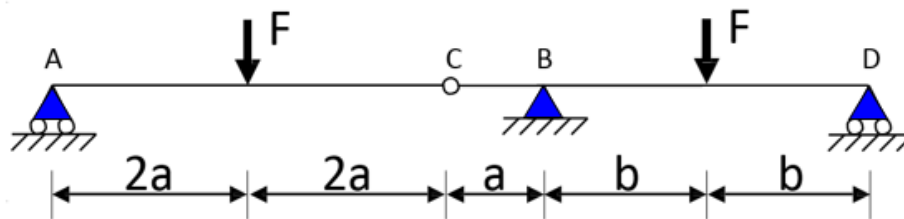
Example 14:



$$\begin{aligned} H_A &= 0 \\ V_A &= -\frac{m}{L} \\ C_A &= m \\ V_B &= \frac{m}{L} \\ H_C &= 0 \\ V_C &= \frac{m}{L} \end{aligned}$$

Example 15:

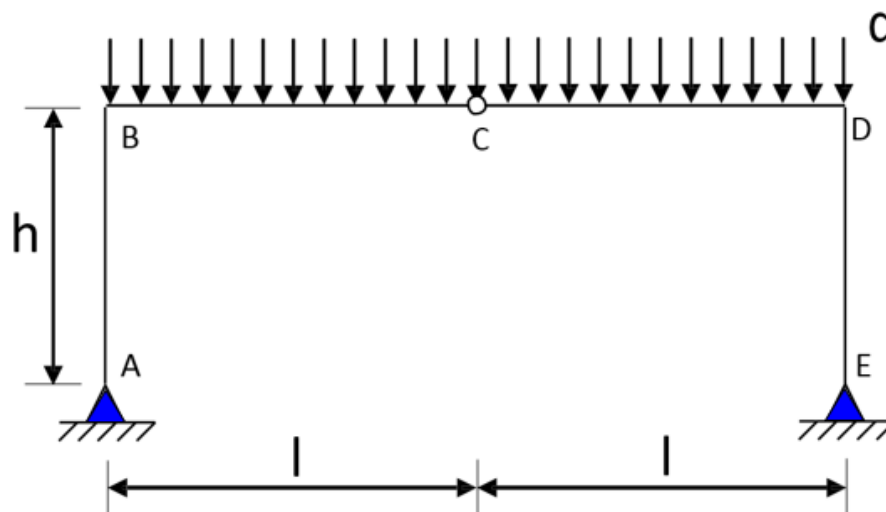
Data: ($a=1.2\text{m}$, $b=2\text{m}$, $F=50\text{N}$).



$$V_A = 25\text{N} \quad V_B = 57.5\text{N} \quad V_C = 25\text{N} \quad H_C = 0 \quad V_D = 17.5\text{N}$$

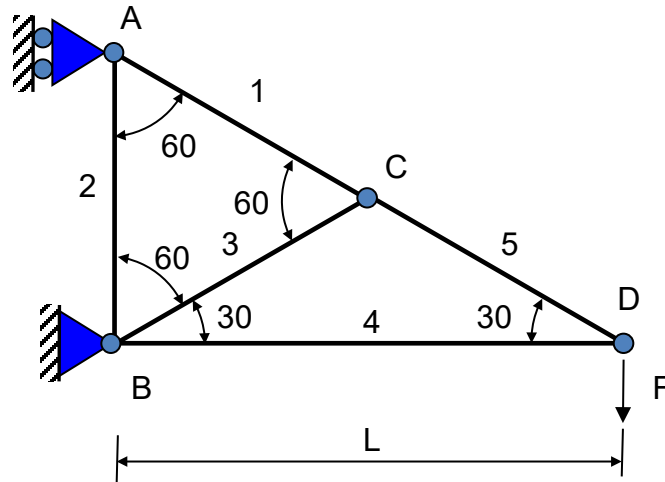
Example 16:

Data: ($l=3\text{m}$, $h=2\text{m}$, $q=20\text{N/m}$),



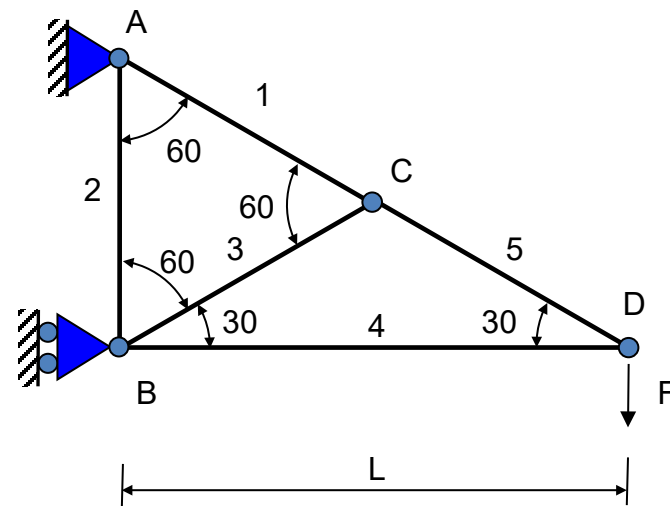
$$V_A = ql \quad H_A = \frac{1}{2} \frac{q}{h} l^2 \quad H_C = \frac{1}{2} \frac{q}{h} l^2 \quad V_C = 0 \quad V_E = ql \quad H_E = \frac{1}{2} \frac{q}{h} l^2$$

Example 17:



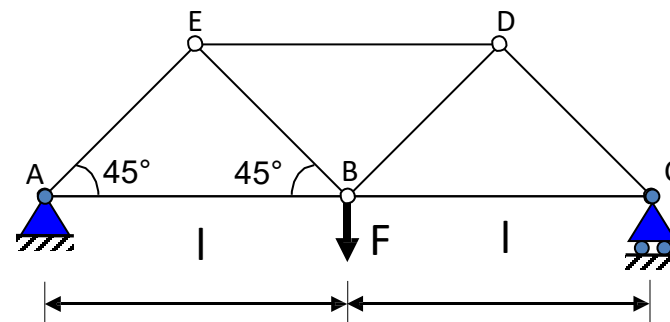
$$V_B = F \quad H_A = -\sqrt{3}F \quad H_B = \sqrt{3}F \quad N_1 = 2F \quad N_2 = -F \quad N_3 = 0 \quad N_4 = -\sqrt{3}F \quad N_5 = 2F$$

Example 18:



$$H_A = -\sqrt{3}F \quad H_B = \sqrt{3}F \quad V_A = -F \quad N_1 = 2F \quad N_2 = 0 \quad N_3 = 0 \quad N_4 = -\sqrt{3}F \quad N_5 = 2F$$

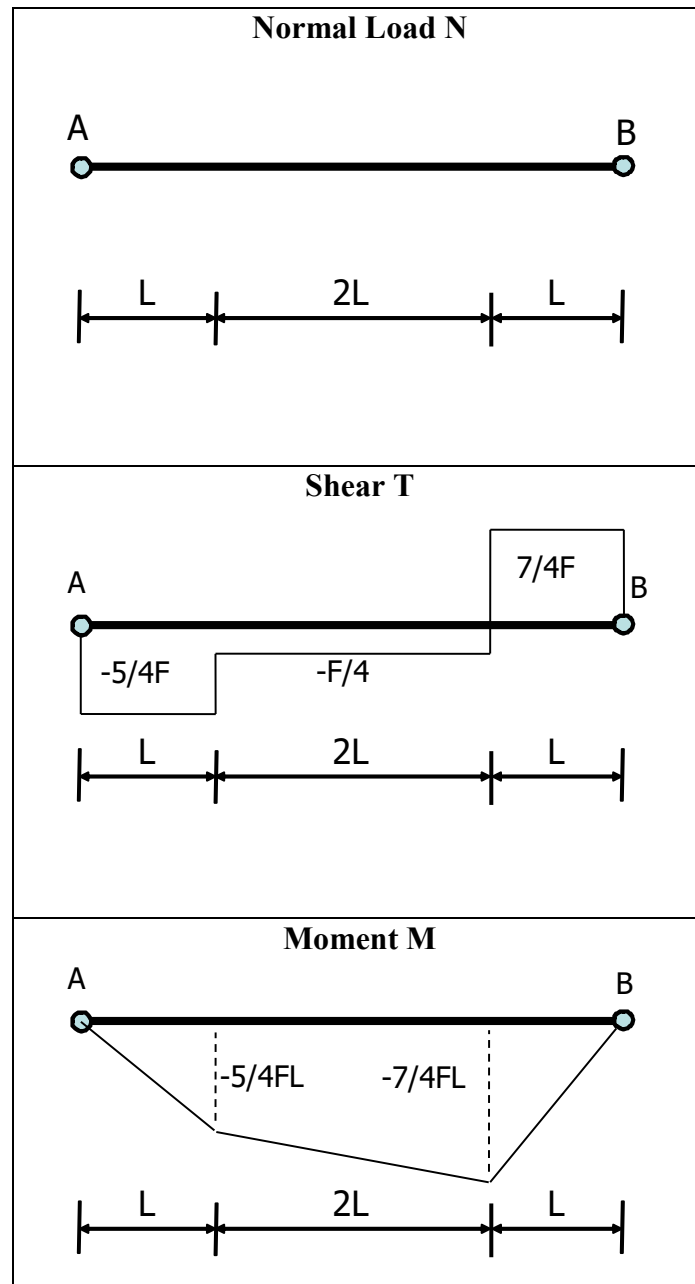
Example 19:



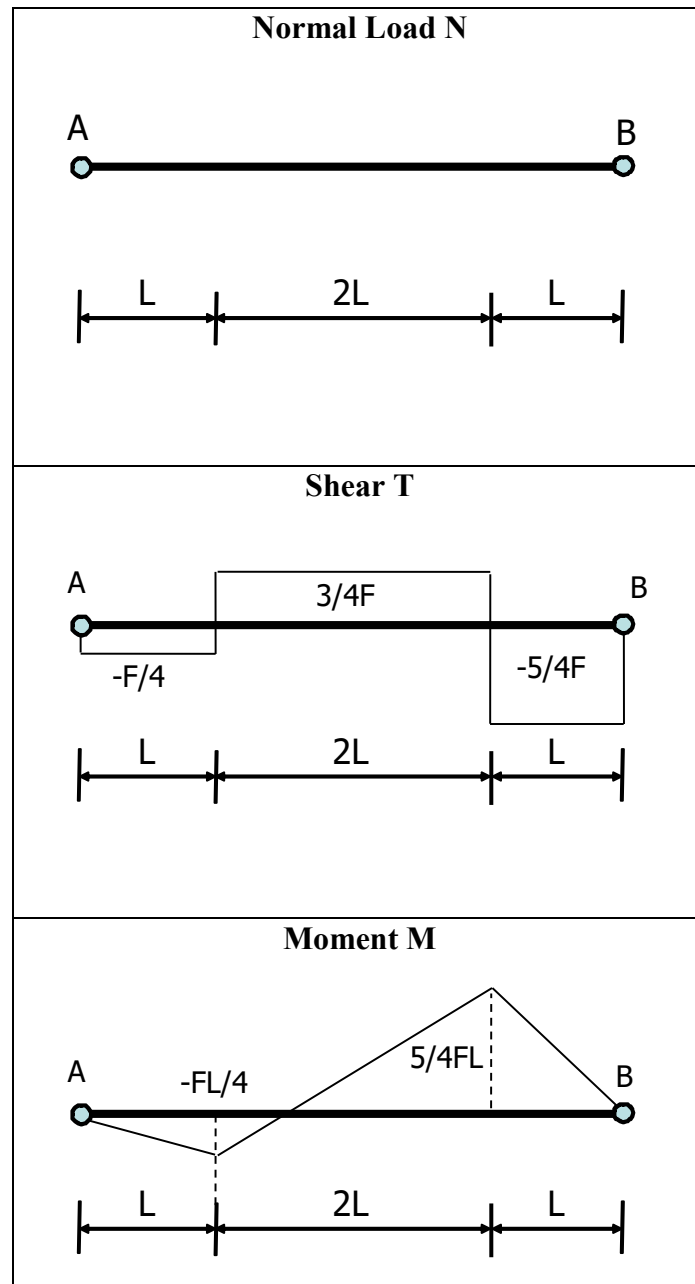
$$H_A = 0 \quad V_A = \frac{F}{2} \quad V_C = \frac{F}{2} \quad N_1 = -\frac{1}{2}F \quad N_2 = \frac{\sqrt{2}}{2}F \quad N_3 = -\frac{\sqrt{2}}{2}F \quad N_4 = F$$

$$N_5 = -\frac{\sqrt{2}}{2}F \quad N_6 = \frac{\sqrt{2}}{2}F \quad N_7 = -\frac{1}{2}F$$

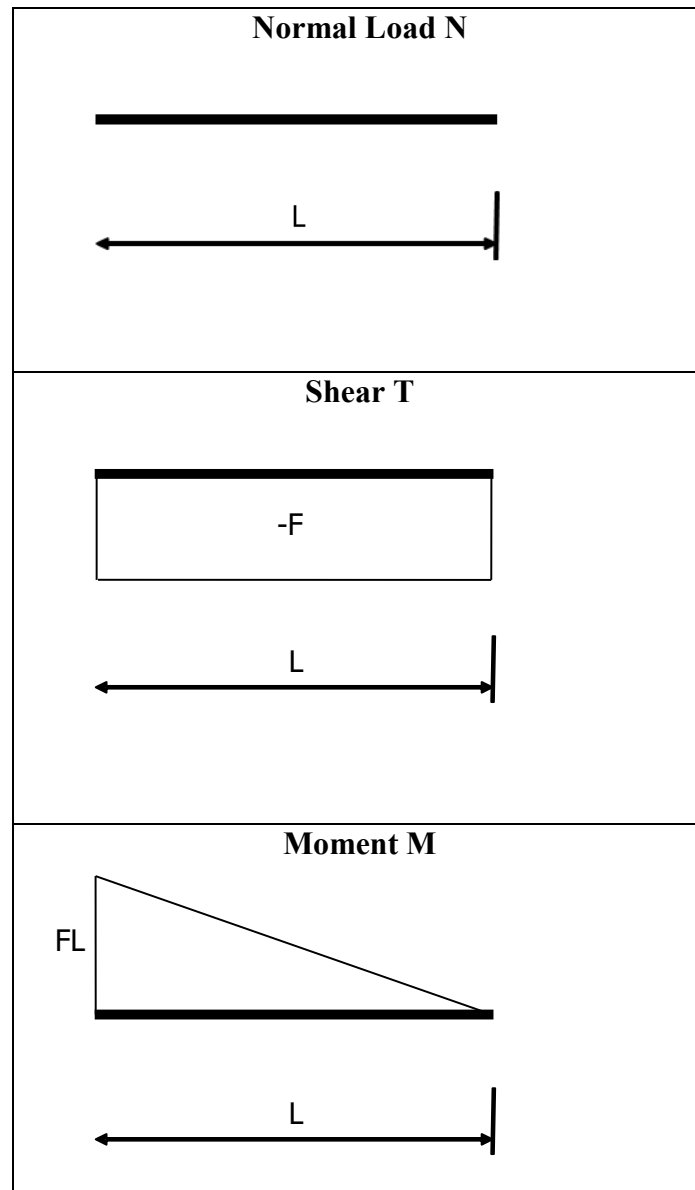
Example 2:



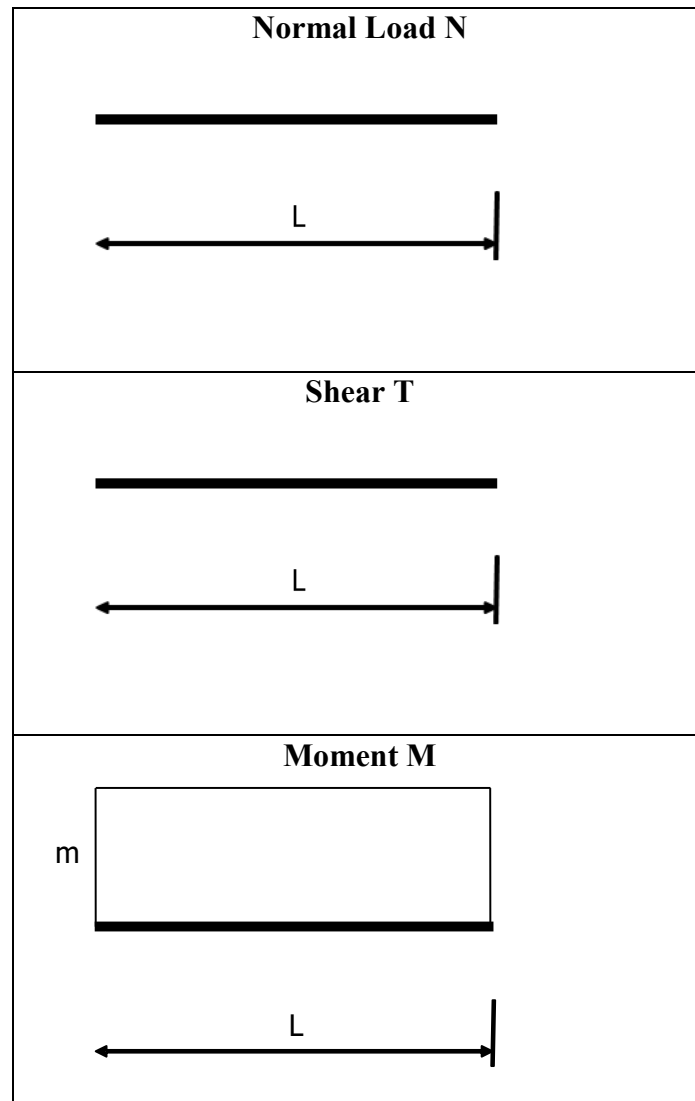
Example 3:



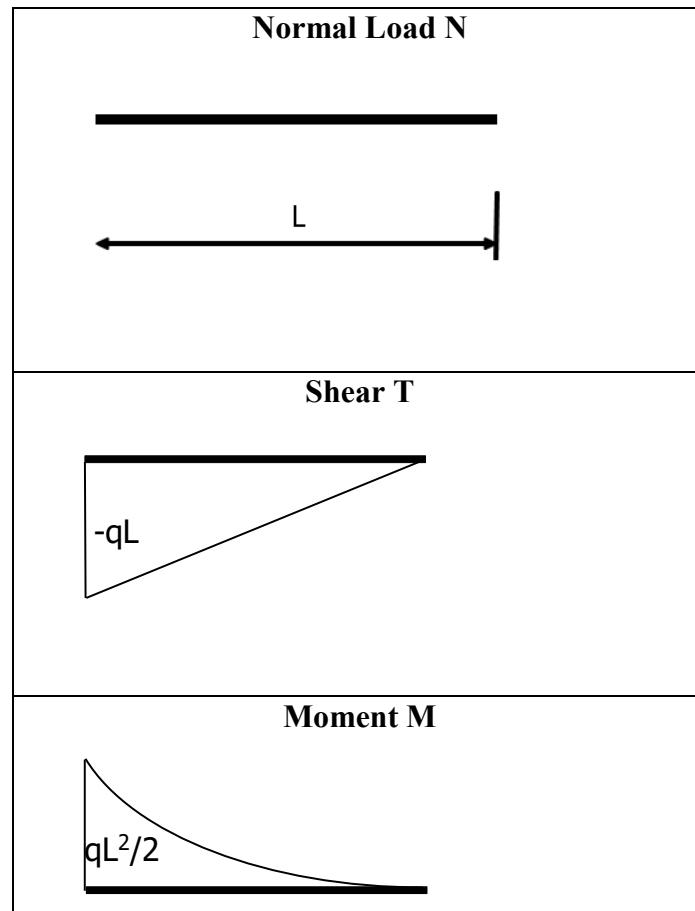
Example 4:



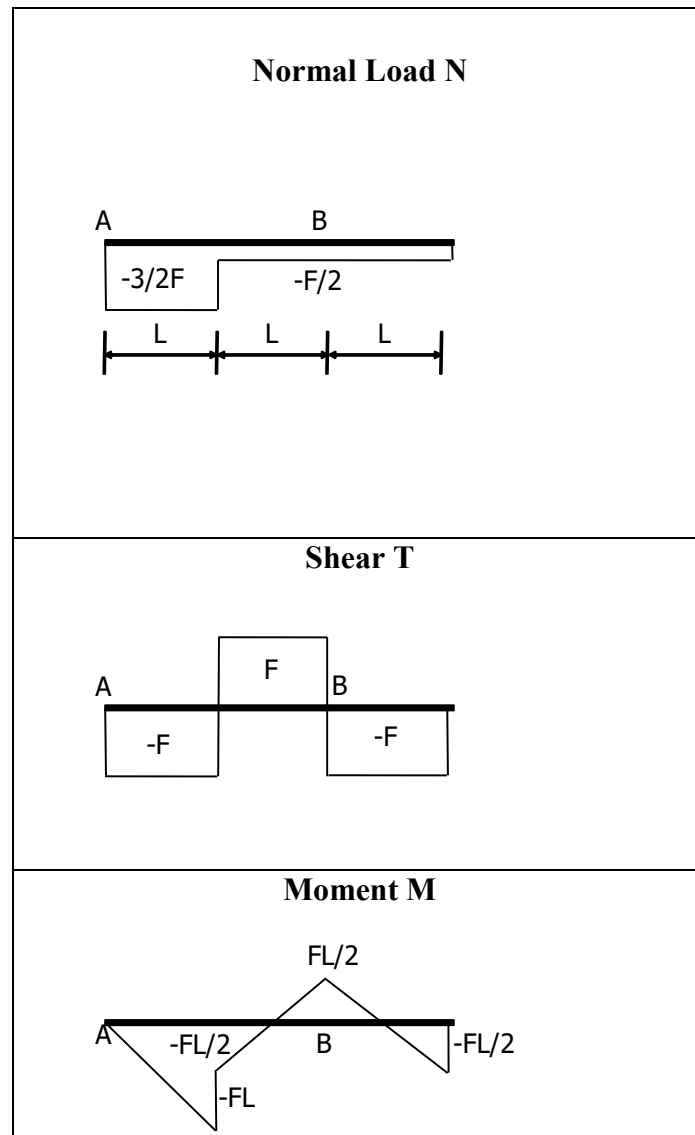
Example 5:



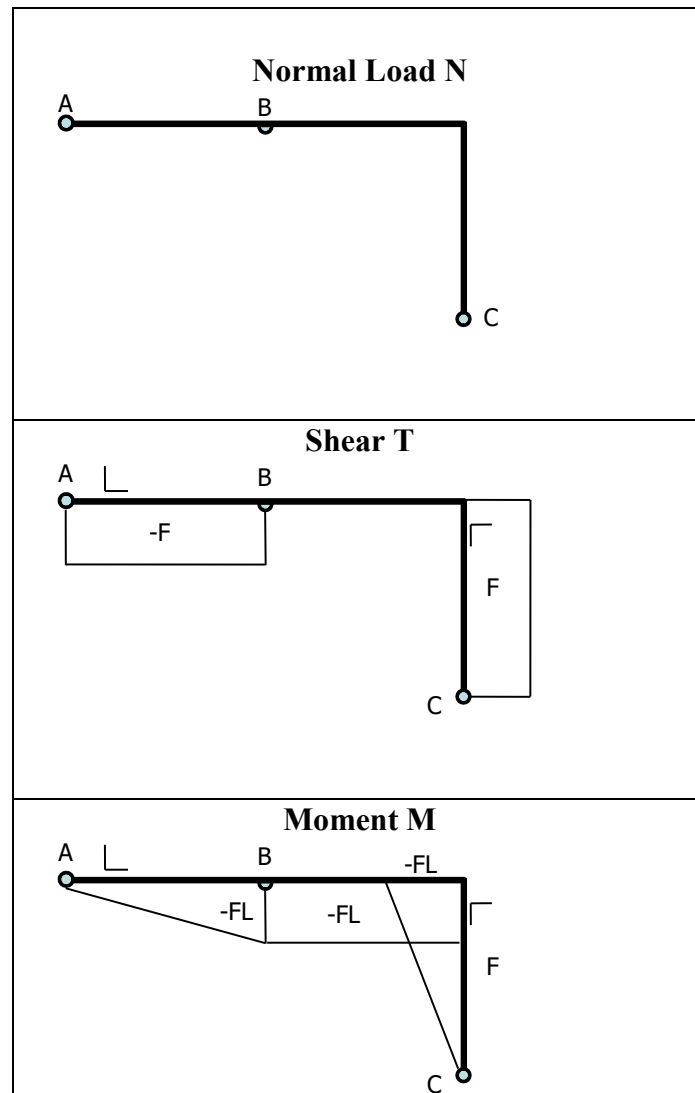
Example 6:



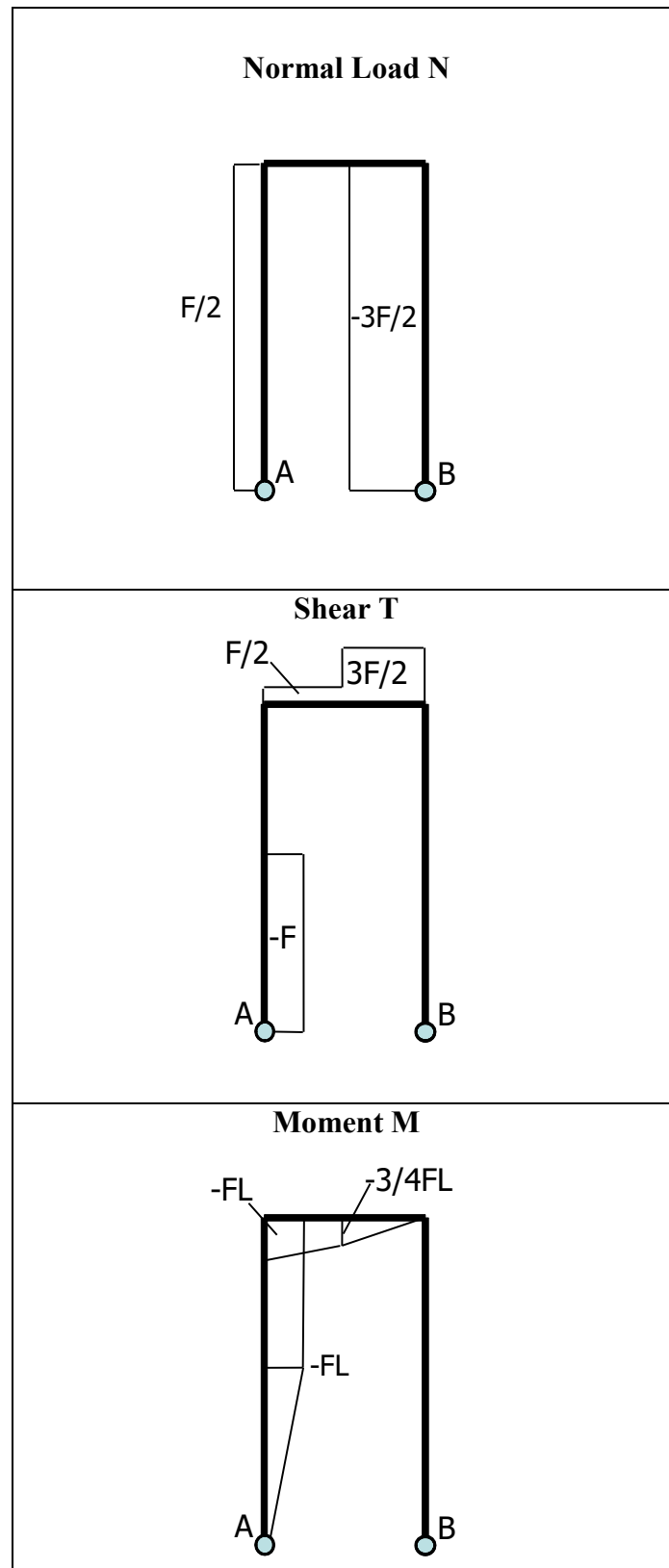
Example 7:



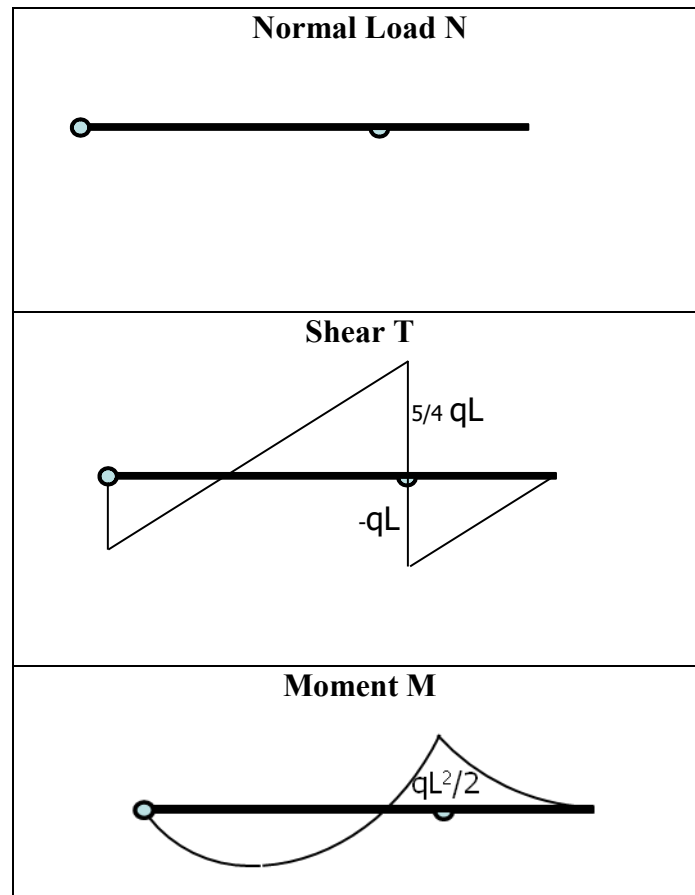
Example 8:



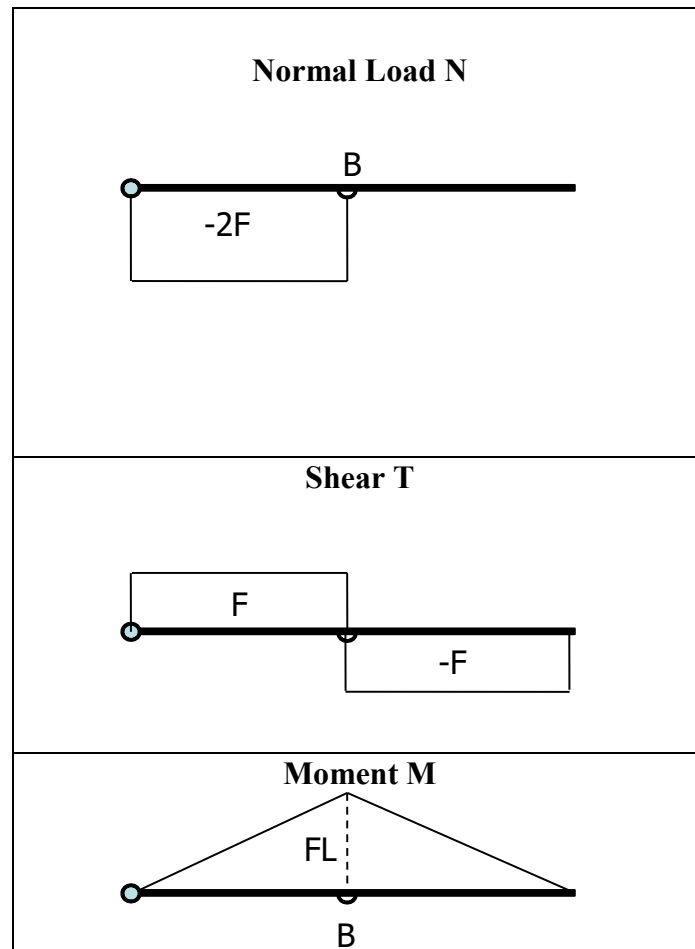
Example 9:



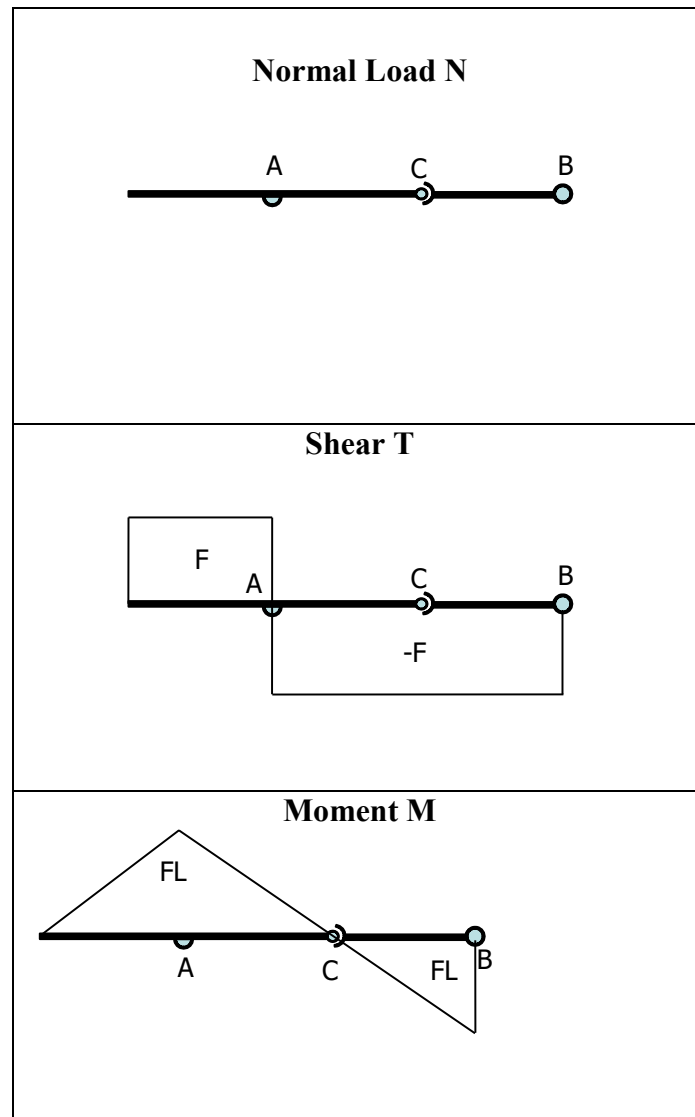
Example 10:



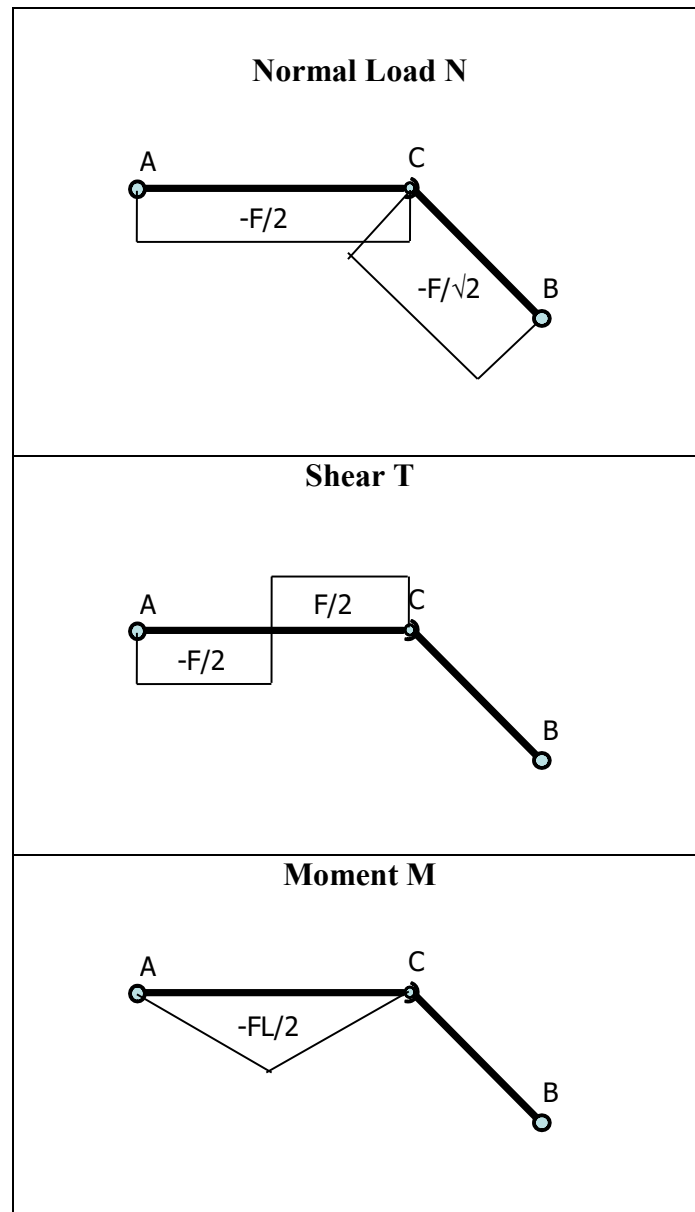
Example 11:



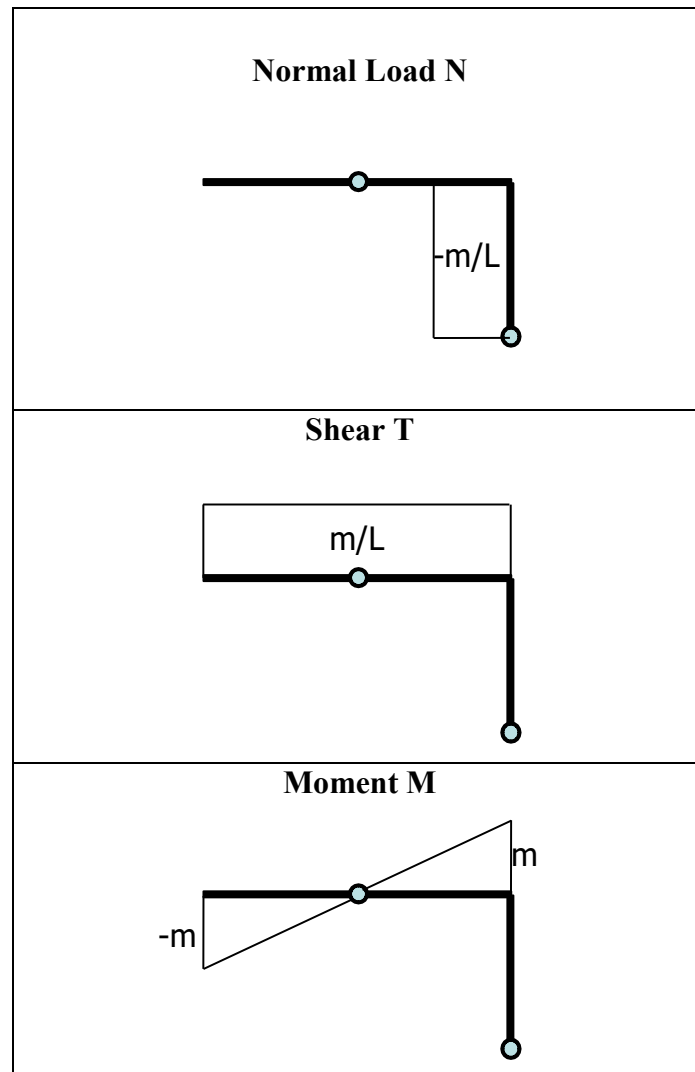
Example 12:



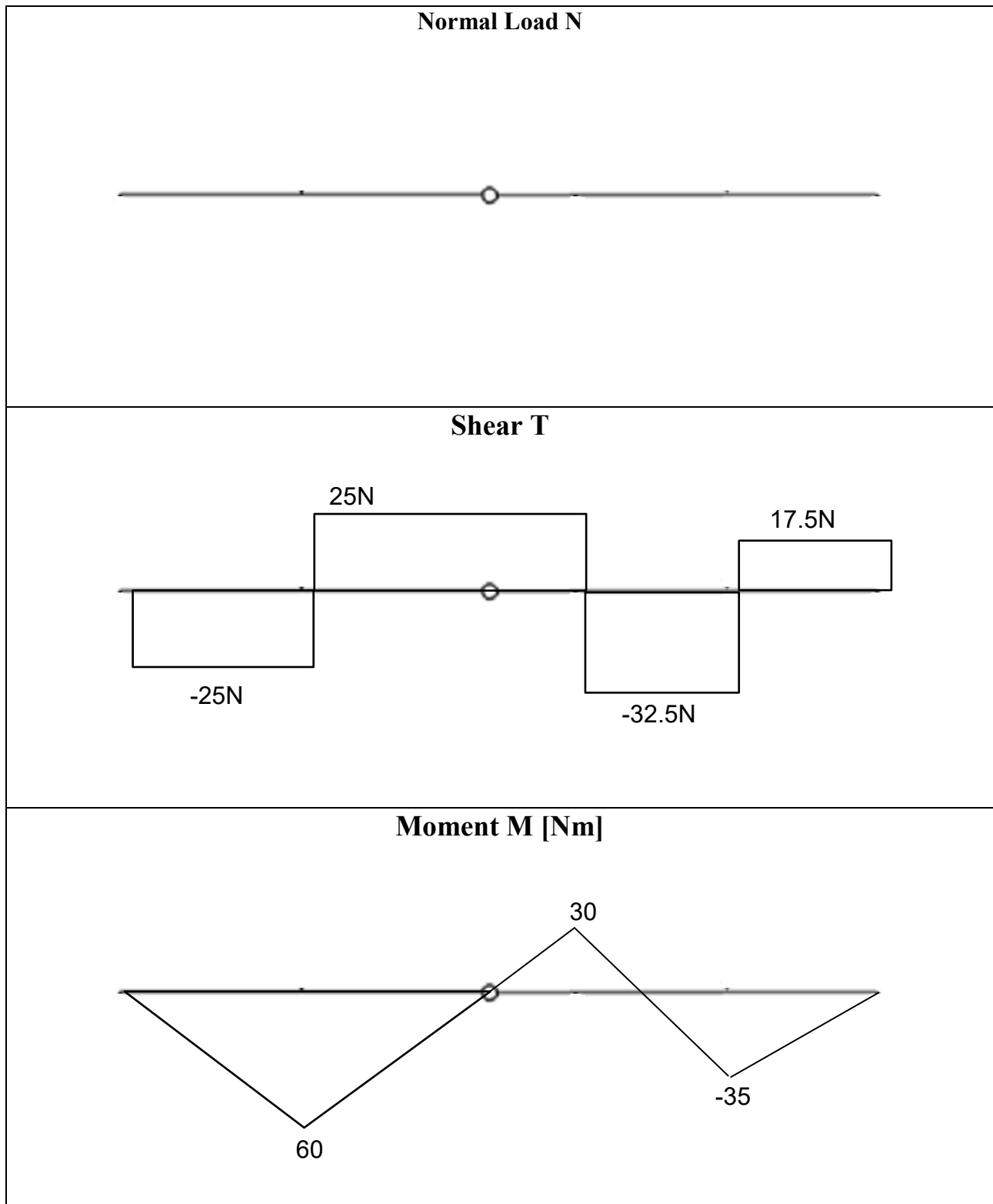
Example 13:



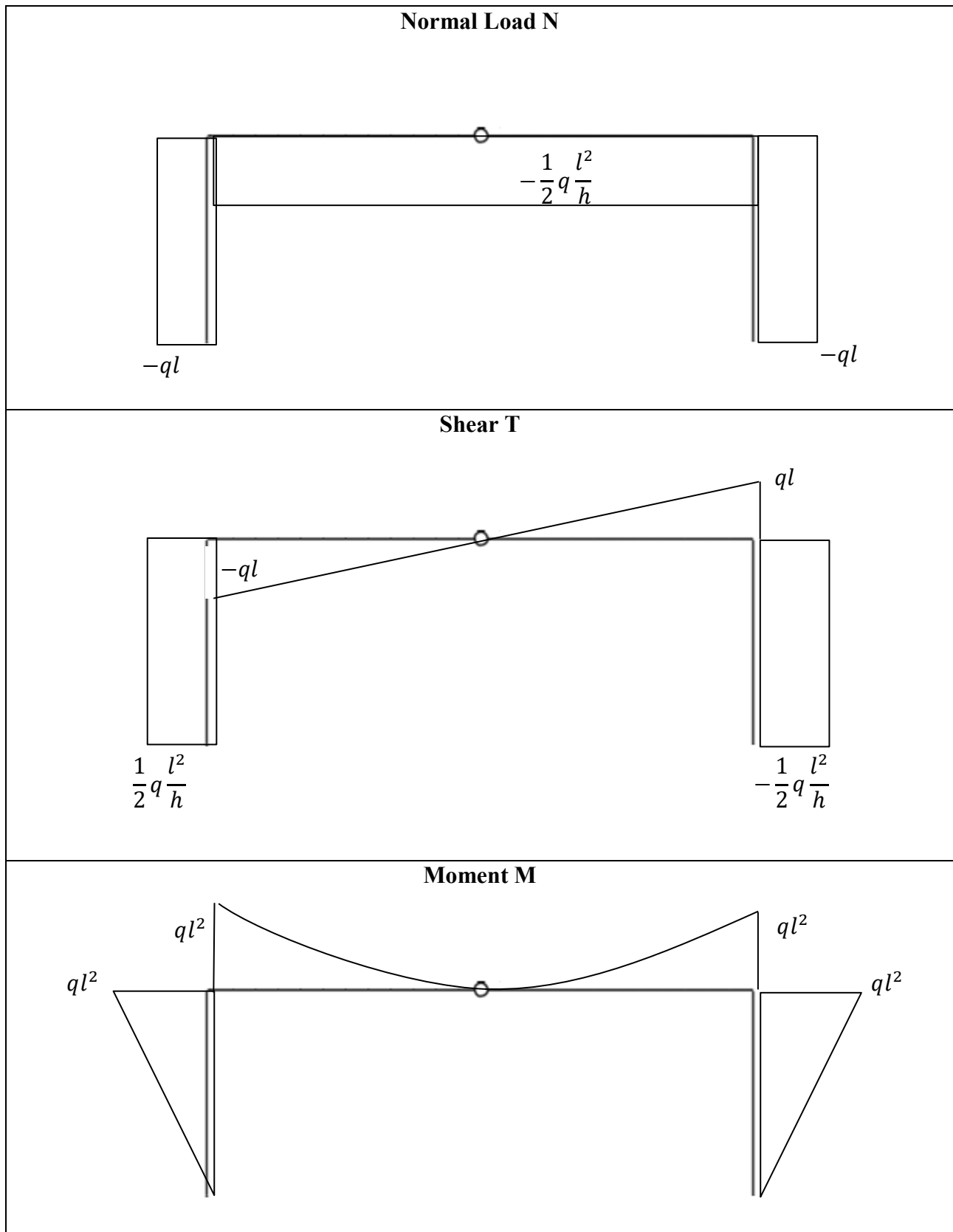
Example 14:



Example 15:

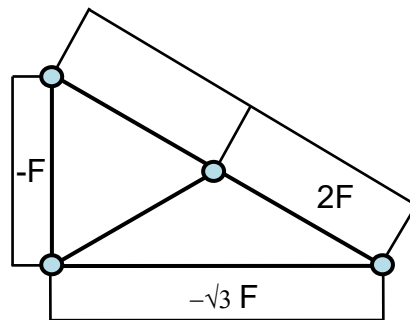


Example 16:



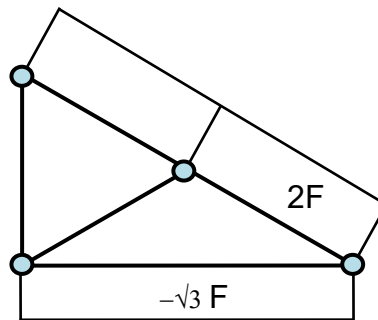
Example 17:

Normal Load N



Example 18:

Normal Load N



Example 19:

Normal Load N

